

Topic 1.1 – DEFINE IoT (Internet of Things)

What is IoT?

IoT (Internet of Things) refers to a network of **physical objects (things)** that are embedded with **sensors, software, and connectivity**, allowing them to **collect, send, and receive data over the internet without human intervention**.

Simple Definition (Exam-friendly):

Internet of Things (IoT) is a system of interconnected devices that can sense, collect, and exchange data using the internet.

Understanding IoT in Simple Words

Normally, the internet connects **people** (using phones, computers).

IoT connects things like:

- Lights
- Fans
- Cars
- Sensors
- Home appliances
- Machines

These things:

1. **Sense data** (temperature, motion, light, humidity, etc.)
 2. **Process data** (using microcontrollers like Raspberry Pi)
 3. **Send data** to the internet or cloud
 4. **Act automatically** based on instructions
-

Basic Example of IoT

Smart Home System

- Temperature sensor detects room temperature
- Data is sent to the internet

- If temperature is high → AC turns ON automatically
- If temperature is low → AC turns OFF

👉 No human action required

Main Components of IoT

IoT system mainly consists of **four parts**:

1. Devices / Things

- Sensors (temperature, humidity, motion)
- Actuators (motors, relays, LEDs)

2. Connectivity

- Wi-Fi
- Bluetooth
- Mobile network
- Internet

3. Data Processing

- Microcontrollers (Raspberry Pi, Arduino)
- Cloud servers

4. User Interface

- Mobile app
 - Web dashboard
 - Alerts / notifications
-

Why IoT is Important

- Automation of daily tasks
- Saves time and energy
- Real-time monitoring
- Remote control of devices
- Improves efficiency and accuracy

Real-Life Examples of IoT

- Smart watches (health monitoring)
 - Smart traffic systems
 - Smart agriculture (soil moisture sensors)
 - Smart cities
 - Industrial automation
-

Relation of IoT with Raspberry Pi & Python

- **Raspberry Pi** acts as the **brain** of IoT devices
- **Python** is used to:
 - Read sensor data
 - Control devices
 - Communicate with servers

That's why this unit combines **Python + Raspberry Pi + IoT**.

Key Points for Exams

- IoT connects **physical objects** to the internet
 - Devices can **sense, process, and act**
 - Reduces human involvement
 - Uses sensors, connectivity, and software
-

Topic 1.2 – APPLICATIONS OF IoT

What are IoT Applications?

IoT applications are the **real-world uses of IoT technology** where connected devices collect data, communicate, and perform actions automatically.

In simple words:

Applications of IoT show **where and how IoT is used in daily life and industries**.

Major Applications of IoT

1. Smart Home

IoT is widely used in homes to make life comfortable and secure.

Examples:

- Smart lights (automatic ON/OFF)
- Smart fans and AC
- Smart door locks
- Smart security cameras

Benefits:

- Saves electricity
 - Remote control using mobile
 - Improves safety
-

2. Smart Healthcare

IoT helps doctors monitor patients remotely.

Examples:

- Heart rate monitoring devices
- Blood pressure sensors
- Smart thermometers
- Wearable fitness trackers

Benefits:

- Real-time patient monitoring
 - Quick medical response
 - Reduces hospital visits
-

3. Smart Agriculture

IoT helps farmers increase crop productivity.

Examples:

- Soil moisture sensors
- Automatic irrigation systems
- Weather monitoring systems

Benefits:

- Saves water
 - Improves crop yield
 - Reduces manual labor
-

4. Smart Cities

IoT improves city infrastructure and services.

Examples:

- Smart street lighting
- Traffic management systems
- Smart parking
- Waste management systems

Benefits:

- Reduces traffic congestion
 - Saves energy
 - Cleaner environment
-

5. Industrial IoT (IIoT)

Used in factories and industries.

Examples:

- Machine health monitoring
- Predictive maintenance
- Automation of manufacturing

Benefits:

- Increases productivity
 - Reduces downtime
 - Improves safety
-

6. Smart Transportation

IoT helps in transportation systems.

Examples:

- GPS vehicle tracking
- Traffic control systems
- Fleet management

Benefits:

- Saves fuel
 - Improves route planning
 - Enhances road safety
-

7. Smart Retail

IoT improves customer experience in shops.

Examples:

- Smart shelves
- Automatic billing
- Inventory tracking

Benefits:

- Better stock management
 - Faster billing
 - Improved customer satisfaction
-

8. Environmental Monitoring

IoT is used to protect the environment.

Examples:

- Air quality monitoring
- Water quality sensors
- Weather monitoring systems

Benefits:

- Pollution control
 - Early warning systems
 - Environmental protection
-

Simple Exam-Oriented Summary

- IoT is used in **homes, hospitals, farms, cities, industries, and transportation**
 - It helps in **automation, monitoring, and control**
 - Improves efficiency and reduces human effort
-

Short Answer (for exams)

Applications of IoT include smart homes, healthcare, agriculture, smart cities, industrial automation, transportation, retail, and environmental monitoring.

Topic 1.3 – ADVANTAGES AND DISADVANTAGES OF IoT

Advantages of IoT**1. Automation**

IoT devices work **automatically without human intervention.**

Example:

Lights turn ON automatically when someone enters a room.

2. Time Saving

IoT reduces manual work and saves time.

Example:

Automatic attendance systems, smart irrigation.

3. Remote Monitoring & Control

Devices can be monitored and controlled **from anywhere using the internet**.

Example:

Checking home security cameras using a mobile phone.

4. Improved Efficiency

IoT systems work accurately and continuously.

Example:

Industrial machines monitored 24/7 to avoid breakdowns.

5. Better Data Collection

IoT devices collect **real-time data**.

Example:

Health monitoring devices track heart rate and temperature.

6. Cost Saving (Long Term)

Although initial cost is high, IoT reduces expenses in the long run.

Example:

Smart energy systems reduce electricity bills.

Disadvantages of IoT

1. Security Issues

IoT devices are connected to the internet, making them **vulnerable to hacking**.

2. Privacy Problems

Personal data collected by devices may be misused.

Example:

Health data leakage.

3. High Initial Cost

Installing IoT systems can be expensive.

4. Internet Dependency

IoT systems **require constant internet connection**.

If the internet fails, the system may stop working.

5. Complexity

IoT systems are complex to design, manage, and maintain.

6. Compatibility Issues

Different devices from different manufacturers may not work together.

Comparison Table (Easy to Remember)

Advantages	Disadvantages
Automation	Security risks
Time saving	Privacy issues
Remote control	High cost
High efficiency	Internet dependency
Real-time data	Complex systems

Exam-Oriented Short Answer

Advantages of IoT include automation, time saving, remote monitoring, efficiency, and real-time data collection.

Disadvantages of IoT include security issues, privacy concerns, high cost, internet dependency, and complexity.

Raspberry Pi 3 is a **small, low-cost single-board computer** developed by the **Raspberry Pi Foundation**, designed to promote learning in computer science and electronics.

Definition

The **Raspberry Pi 3** is a credit-card-sized computer that can run a full operating system (usually Linux), connect to peripherals like a keyboard, mouse, and monitor, and be used for programming, electronics projects, and general computing tasks.

Key features (Raspberry Pi 3 Model B)

- **CPU:** Quad-core 64-bit ARM Cortex-A53, 1.2 GHz
- **RAM:** 1 GB
- **Connectivity:** Built-in Wi-Fi (802.11n) and Bluetooth 4.1
- **Ports:** HDMI, USB, Ethernet, audio jack, GPIO pins
- **Storage:** microSD card
- **Power:** 5V via micro-USB

Common uses

- Learning programming (Python, Scratch, etc.)
- Electronics and robotics projects
- Media centers
- Home automation
- Small servers and IoT applications

In short, the **Raspberry Pi 3** is a versatile, affordable computer widely used for **education, experimentation, and DIY projects**.

Excellent 👍

Now let's move to the next topic.

Topic 1.4 – COMPONENTS OF RASPBERRY PI-3

What is Raspberry Pi?

Raspberry Pi 3 is a **small, low-cost, single-board computer** used for:

- Learning programming (Python)
- IoT projects
- Embedded systems
- Automation

It works like a normal computer but on a small board.

Main Components of Raspberry Pi-3

1. Processor (CPU)

- **Type:** Quad-core ARM Cortex-A53
 - **Speed:** 1.2 GHz
 - Acts as the **brain** of Raspberry Pi
 - Executes programs and controls all operations
-

2. RAM (Memory)

- **1 GB LPDDR2 RAM**
 - Stores data temporarily while programs are running
 - More RAM = better multitasking
-

3. USB Ports

- **4 × USB 2.0 ports**
 - Used to connect:
 - Keyboard
 - Mouse
 - Pen drive
 - USB Wi-Fi adapters
-

4. HDMI Port

- Used to connect Raspberry Pi to a **monitor or TV**

- Supports video and audio output
-

5. Ethernet Port

- Used for **wired internet connection**
 - Provides stable network access
-

6. Built-in Wi-Fi & Bluetooth

- **Wi-Fi:** 802.11 b/g/n
- **Bluetooth:** 4.1

Allows wireless internet and device connectivity.

7. GPIO Pins (40-pin Header)

- Used to connect:
 - Sensors
 - LEDs
 - Motors
 - Relays

Very important for **IoT projects**.

8. Micro SD Card Slot

- Used to insert **micro SD card**
 - SD card acts as:
 - Hard disk
 - Stores OS, programs, and files
-

9. Power Supply Port

- **Micro-USB port**
- Requires **5V, 2.5A** power supply

10. Camera Interface (CSI)

- Used to connect **Raspberry Pi Camera Module**
- Used in:
 - Surveillance
 - Face recognition
 - Image processing

11. Display Interface (DSI)

- Used to connect **LCD touchscreen display**

12. Audio Jack

- **3.5 mm jack**
- Used for headphones or speakers

Simple Diagram Understanding (Conceptual)

- CPU → Processing
- RAM → Temporary memory
- SD Card → Storage
- GPIO → Hardware interfacing
- Wi-Fi → Internet connectivity

Why Raspberry Pi-3 is Used in IoT

- Small size
 - Low power consumption
 - Supports Python
 - Easy hardware interfacing
-

Exam-Friendly Short Answer

Raspberry Pi-3 components include CPU, RAM, USB ports, HDMI port, Ethernet, Wi-Fi, Bluetooth, GPIO pins, SD card slot, power supply, camera interface, and audio jack.

Topic 1.5 – GENERAL PURPOSE INPUT AND OUTPUT PINS (GPIO) ON RASPBERRY PI

What are GPIO Pins?

GPIO (General Purpose Input Output) pins are **special pins on Raspberry Pi** used to:

- **Take input** from external devices (sensors, switches)
- **Give output** to external devices (LEDs, motors, buzzers)

They help Raspberry Pi **communicate with the outside world**.

GPIO Pin Header

- Raspberry Pi has **40 GPIO pins**
 - Arranged in **2 rows of 20 pins**
 - Each pin has a specific function
-

Types of GPIO Pins

1. Power Pins

Used to supply power to external components.

Voltage Purpose

5V	Power to motors, relays
3.3V	Power to sensors
GND	Ground connection

2. GPIO Pins (Digital Pins)

- Used as **Input or Output**
- Controlled using **Python programs**

Example:

- Input → Push button
 - Output → LED
-

3. Special Function Pins

These pins perform advanced communication.

Pin Type Function

UART	Serial communication
I2C	Communication with sensors
SPI	High-speed data transfer
PWM	Speed control (motors, LEDs)

GPIO Pin Numbering Systems

1. BOARD Numbering

- Uses **physical pin numbers**
- Example: Pin 7, Pin 11

2. BCM Numbering

- Uses **Broadcom chip numbers**
- Example: GPIO4, GPIO17

👉 **BCM numbering is mostly used in Python programs**

Input and Output Explained

GPIO as Input

- Reads data from sensors
- Reads button state (pressed / not pressed)

Example:

Temperature sensor sending data to Raspberry Pi

GPIO as Output

- Sends signals to devices
- Turns devices ON or OFF

Example:

Turning ON an LED using Python

Important Safety Rule ⚠

- GPIO pins work at **3.3V**
 - Applying **5V directly** to GPIO pins can **damage Raspberry Pi**
-

Why GPIO is Important for IoT

- Connects Raspberry Pi to real-world devices
 - Enables automation
 - Allows hardware control using Python
-

Exam-Oriented Short Answer

GPIO pins are used to interface Raspberry Pi with external devices by acting as input or output pins for sensors and actuators.

If you are **done with GPIO pins**, reply:

Excellent 👍

Now let's move to the next topic.

Topic 1.6 – BUILD A PC USING RASPBERRY PI-3

What Does “Build a PC using Raspberry Pi-3” Mean?

It means using **Raspberry Pi-3 as a complete personal computer** by connecting required peripherals such as:

- Monitor
- Keyboard
- Mouse
- Power supply
- SD card with OS

Raspberry Pi can perform basic PC tasks like:

- Programming (Python)
 - Browsing internet
 - Playing videos
 - Writing documents
-

Requirements to Build a PC

1. Raspberry Pi-3 Board

Acts as the **CPU + motherboard**.

2. Micro SD Card

- Minimum **8 GB (recommended 16 GB)**
 - Contains **Operating System (OS)** like:
 - Raspbian (Raspberry Pi OS)
-

3. Power Supply

- **5V, 2.5A Micro-USB power adapter**
 - Provides power to Raspberry Pi
-

4. Display Unit

- Monitor or TV
 - Connected using **HDMI cable**
-

5. Input Devices

- USB Keyboard
 - USB Mouse
-

6. Internet Connection (Optional)

- Wi-Fi or Ethernet cable
-

Steps to Build a PC Using Raspberry Pi-3

Step 1: Install Operating System

- Download **Raspberry Pi OS**
 - Write OS to micro SD card using tools like Raspberry Pi Imager
-

Step 2: Insert SD Card

- Insert the micro SD card into Raspberry Pi SD slot
-

Step 3: Connect Peripherals

- Connect keyboard and mouse to USB ports
 - Connect monitor using HDMI cable
-

Step 4: Power ON

- Connect power supply
 - Raspberry Pi boots automatically
-

Step 5: Configure System

- Set language, time, Wi-Fi
 - Desktop screen appears
-

Functions of Raspberry Pi PC

- Python programming
 - Internet browsing
 - Office applications
 - Media playback
 - IoT development
-

Advantages of Raspberry Pi PC

- Low cost
 - Low power consumption
 - Portable
 - Ideal for students
-

Limitations

- Not suitable for heavy gaming
 - Lower performance compared to desktop PCs
-

Exam-Oriented Short Answer

A PC can be built using Raspberry Pi-3 by connecting a monitor, keyboard, mouse, SD card with OS, and power supply.

Next Topic – INSTALL OS TO RASPBERRY PI

What Does “Install OS to Raspberry Pi” Mean?

Installing an OS means **loading an operating system** (like Windows or Linux) onto the **micro SD card**, which allows Raspberry Pi to work like a computer.

👉 Raspberry Pi **does not have a hard disk**, so the **SD card acts as its storage**.

Common Operating System for Raspberry Pi

- **Raspberry Pi OS (Raspbian)** – *Most commonly used*
- Ubuntu
- Windows IoT Core (limited use)

 **In exams, Raspberry Pi OS is the expected answer.**

Requirements to Install OS

1. Raspberry Pi-3
 2. Micro SD Card (8GB or more)
 3. SD Card Reader
 4. Computer / Laptop
 5. Internet Connection
 6. Raspberry Pi Imager software
-

Steps to Install OS on Raspberry Pi

Step 1: Download Raspberry Pi Imager

- Official tool used to install OS on SD card
-

Step 2: Insert SD Card

- Insert micro SD card into **card reader**
 - Connect it to your computer
-

Step 3: Open Raspberry Pi Imager

Select:

- **Operating System** → Raspberry Pi OS
 - **Storage** → Your SD card
-

Step 4: Write OS to SD Card

- Click **WRITE**

- OS files are copied to SD card
 - Process may take a few minutes
-

Step 5: Insert SD Card into Raspberry Pi

- Remove SD card from computer
 - Insert into Raspberry Pi SD card slot
-

Step 6: Power ON Raspberry Pi

- Connect monitor, keyboard, mouse
 - Plug in power supply
 - Raspberry Pi boots into OS
-

First Boot Configuration

On first boot:

- Select language
 - Set time zone
 - Connect Wi-Fi
 - Change password (optional)
-

Why OS Installation is Important

- Without OS, Raspberry Pi **cannot function**
 - OS manages:
 - Hardware
 - Memory
 - Files
 - Programs (Python)
-

Advantages of Raspberry Pi OS

- Free and open source
 - Optimized for Raspberry Pi
 - Comes with Python pre-installed
 - Beginner-friendly
-

Exam-Oriented Short Answer

Installing OS on Raspberry Pi involves writing Raspberry Pi OS to a micro SD card using Raspberry Pi Imager and booting the Raspberry Pi using that SD card.

1.8 FAMILIARIZE TO PYTHON PROGRAMMING LANGUAGE

No extra topics. No mixing with later headings.

1.8 FAMILIARIZE TO PYTHON PROGRAMMING LANGUAGE

Meaning

To *familiarize* means **to get introduced and become comfortable** with Python programming.

This topic explains:

- What Python is
 - How Python programs look
 - Basic way of writing and understanding Python code
 - How Python works in general
-

What is Python Programming Language?

Python is a **high-level, interpreted programming language** that is:

- Easy to learn
- Easy to read
- Easy to write

Python is widely used for:

- Raspberry Pi programming
- IoT applications

- Automation
-

Characteristics of Python (Basic Introduction)

1. Simple and Readable Syntax

Python uses simple English-like statements.

Example:

```
print("Hello Python")
```

This makes Python easy for beginners.

2. Interpreted Language

- Python code is executed **line by line**
- No need to compile the program before running

This helps in easy error detection.

3. High-Level Language

- Programmer does not need to worry about hardware details
 - Memory management is handled automatically
-

4. Case Sensitive Language

Python differentiates between uppercase and lowercase letters.

Example:

```
count = 5
```

```
Count = 10
```

Both variables are **different**.

5. Uses Indentation

Python uses **indentation (spaces)** to define blocks of code instead of { }.

Example:

if 5 > 3:

```
    print("True")
```

Indentation is **mandatory** in Python.

Writing a Simple Python Program

Example Program

```
print("Welcome to Python Programming")
```

Explanation

- `print()` is a built-in function
 - Displays output on the screen
-

How Python Programs Work

1. Programmer writes Python code
 2. Python interpreter reads the code
 3. Code is executed line by line
 4. Output is displayed
-

Python in Raspberry Pi

- Python comes **pre-installed** in Raspberry Pi OS
 - Used to:
 - Control GPIO pins
 - Read sensor data
 - Build IoT applications
-

Why Python is Suitable for Beginners

- Simple syntax
- Less code
- Easy to understand

- Fast learning
-

Exam-Oriented Answer (Very Important)

Familiarize to Python programming language means understanding the basic concepts of Python such as its simple syntax, interpreted nature, indentation, case sensitivity, and basic program execution.

Good 

Now we move **exactly to the next topic in your lesson.**

1.9 DIFFERENT VERSIONS OF PYTHON

What are Python Versions?

Python versions refer to the **different releases of the Python programming language** developed over time.

Each version improves:

- Performance
 - Features
 - Security
 - Libraries
-

Major Python Versions

Python is mainly divided into **two major versions**:

1. Python 2

- Released in **2000**
- Older version of Python
- Simple and powerful
- **Now discontinued (not supported)**

 Python 2 is **only for theoretical understanding.**

2. Python 3

- Released in **2008**
- Latest and most widely used version
- Improved features and performance
- **Officially supported**

 Raspberry Pi uses **Python 3** by default.

Differences Between Python 2 and Python 3 (Basic)

Feature	Python 2	Python 3
Print statement	<code>print "Hi"</code>	<code>print("Hi")</code>
Unicode support	Limited	Full support
Division	Integer division	Float division
Support status	Discontinued	Actively supported

Why Python 3 is Preferred

- Better syntax
 - Improved security
 - Better library support
 - Future-ready
-

Python Versions Used in Raspberry Pi

- Python 3 is **pre-installed**
 - Used for:
 - GPIO programming
 - IoT applications
 - Automation
-

Exam-Oriented Short Answer

Python has different versions, mainly Python 2 and Python 3. Python 2 is discontinued, while Python 3 is the latest and widely used version for current applications.

Great 👍

Now we continue **exactly as per your lesson outline**.

1.10 PYTHON VS JAVA

Introduction

Python and Java are both **high-level programming languages**, but they differ in:

- Syntax
- Execution
- Usage
- Complexity

This topic compares Python and Java to understand their differences.

Comparison Between Python and Java

Feature	Python	Java
Syntax	Simple and readable	Complex and lengthy
Code length	Short programs	Longer programs
Compilation	Interpreted	Compiled + Interpreted
Speed	Slower than Java	Faster than Python
Ease of learning	Very easy	Moderately difficult
Usage	IoT, AI, automation	Enterprise, Android apps
Memory management	Automatic	Automatic (JVM)

Example Comparison

Python Code

```
print("Hello World")
```

Java Code

```
class Hello {  
    public static void main(String args[]) {  
        System.out.println("Hello World");  
    }  
}
```

👉 Python code is **shorter and simpler**.

Why Python is Preferred for Beginners

- Easy syntax
 - Less code
 - Faster learning
 - Ideal for Raspberry Pi and IoT
-

Why Java is Still Used

- High performance
 - Platform independent (JVM)
 - Strong security
 - Used in large enterprise applications
-

Exam-Oriented Short Answer

Python is simpler and easier to learn compared to Java, while Java is faster and widely used in enterprise applications.

✓ **STOP POINT**

Excellent 👍

Now we move to the **next topic exactly as per your lesson.**

1.11 FEATURES OF PYTHON

What are Features of Python?

Features of Python describe the **special characteristics** that make Python popular and widely used.

Main Features of Python

1. Simple and Easy to Learn

- Uses English-like syntax
 - Easy for beginners
-

2. Interpreted Language

- Code is executed line by line
 - No need to compile before execution
-

3. High-Level Language

- No need to manage memory
 - Focus on logic rather than hardware
-

4. Open Source

- Free to use
 - Source code is available
-

5. Portable (Platform Independent)

- Runs on Windows, Linux, macOS, Raspberry Pi

6. Object-Oriented Programming Support

- Supports classes and objects
- Encourages code reuse

7. Large Standard Library

- Provides built-in modules and functions
- Reduces coding effort

8. Dynamically Typed

- No need to declare data types
- Python identifies data type automatically

9. Extensible and Embedded

- Can be integrated with other languages like C and C++

10. Suitable for IoT and Raspberry Pi

- Easy GPIO control
- Hardware-friendly

Exam-Oriented Short Answer

Python features include simple syntax, interpreted nature, high-level language, portability, open-source availability, object-oriented support, large libraries, and dynamic typing.

Great 👍

Now we move to the **next topic exactly as per your lesson.**

1.12 SETTING THE EXECUTION ENVIRONMENT FOR PYTHON

What is Execution Environment?

The **execution environment** is the **setup required to write, run, and test Python programs**.

It includes:

- Python interpreter
 - Editor or IDE
 - Operating system
-

Requirements to Set Python Execution Environment

1. Operating System

Python can run on:

- Windows
 - Linux
 - macOS
 - Raspberry Pi OS
-

2. Python Interpreter

- Required to execute Python programs
- Converts Python code into machine-readable instructions

✂ Python 3 is commonly used.

3. Text Editor / IDE

Used to write Python programs.

Common options:

- IDLE (comes with Python)
 - Thonny (default in Raspberry Pi)
 - VS Code
 - PyCharm
-

Setting Python Environment in Windows

Steps

1. Download Python from official website
2. Install Python
3. Check installation using:

```
python --version
```

4. Open IDLE or Command Prompt
 5. Run Python programs
-

Setting Python Environment in Raspberry Pi

Steps

1. Raspberry Pi OS comes with Python pre-installed
 2. Open **Thonny Python IDE**
 3. Write Python code
 4. Run the program
-

Python Execution Modes

1. Interactive Mode

- Commands are executed immediately
- Used for testing

Example:

```
>>> print("Hello")
```

2. Script Mode

- Python programs saved as `.py` files

- Executed as a whole

Example:

```
hello.py
```

Exam-Oriented Short Answer

Setting the execution environment for Python involves installing the Python interpreter and using an editor or IDE to write and execute Python programs.

Perfect 👍

Now we proceed **exactly as per your lesson**.

1.13 DIFFERENT WAYS TO EXECUTE PYTHON SCRIPT

What Does Executing a Python Script Mean?

Executing a Python script means **running a Python program** so that the instructions written in the program are performed and output is produced.

Different Ways to Execute Python Script

1. Interactive Mode

- Python commands are executed **one by one**
- Used for quick testing

Example:

```
>>> print("Hello Python")
```

✂ No file is saved.

2. Script Mode

- Program is written in a file with `.py` extension
- Entire program is executed at once

Steps:

1. Write program in editor
 2. Save as `example.py`
 3. Run the file
-

3. Using IDLE

- Open IDLE
 - Write code in editor window
 - Click **Run** → **Run Module**
-

4. Using Command Prompt / Terminal

Run using command:

```
python filename.py
```

5. Using IDE (Thonny / VS Code)

- Write program
- Click **Run** button

Used mostly in Raspberry Pi.

Difference Between Interactive and Script Mode

Interactive Mode	Script Mode
Executes line by line	Executes entire program
No file required	File required
Used for testing	Used for full programs

Exam-Oriented Short Answer

Python scripts can be executed using interactive mode, script mode, IDLE, command prompt, or Python IDEs such as Thonny.
